Practice Exercises: Differential Equations

1. Introduction to Differential Equations

1. Determine the order and linearity of the following differential equations:
   a) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 2y = 0$

   b) $xy'' + \sin(x)y' + e^y = 0$

   c) $y' + xy = x^2$

   d) $\frac{d^3y}{dx^3} + \cos(y') = x^2$
2. Classify each of the following equations as an **ODE** or **PDE**:
   a) $\frac{\partial u}{\partial t} + \frac{\partial^2 u}{\partial x^2} = 0$

   b) $y'' - y' + y = x$

   c) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

   d) $y' + \sin(y) = x^2$

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2. First-Order Ordinary Differential Equations

3. Solve the following separable equations:
   a) $\frac{dy}{dx} = xy$

   b) $\frac{dy}{dx} = \frac{x}{y}$

   c) $y' = e^x y$

   d) $(y^2 + 1)dy = xdx$
4. Solve the homogeneous differential equations:
   a) $y' = \frac{x+y}{x-y}$

   b) $y' = \frac{x^2+y^2}{xy}$
5. Solve the first-order linear equations using the integrating factor method:
   a) $y' + 2y = e^{-x}$

   b) $xy' + y = x^2$

   c) $y' + \frac{y}{x} = \sin x$
6. Determine whether the following equations are **exact**, and if so, solve them:
   a) $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$

b) $(y + 3x^2y^2)dx + (x + 2x^3y)dy = 0$

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3. Applications of First-Order Differential Equations

7. **Exponential Growth & Decay:** A bacteria culture starts with 500 cells and grows at a rate proportional to its population. After 2 hours, there are 2000 cells.
   - Find the population function $P(t)$.

   - Determine how many cells will be present after 5 hours.
8. **Newton's Law of Cooling:** A cup of coffee at $90°C$ is left in a room at $20°C$. After 5 minutes, the temperature drops to $70°C$.
   - Find the cooling equation.

   - Determine the temperature after 10 minutes.
9. **Falling Object with Air Resistance:** A 5 kg object is dropped from rest, experiencing air resistance proportional to its velocity, given by $m\frac{dv}{dt} = mg - kv$ with $k = 2$ Ns/m.
   - Find the velocity function $v(t)$.

   - Determine the terminal velocity.